



CODECLASH

Problem Statements

I. Agriculture, FoodTech & Rural Development

1. Analyzing data derived from the Govt. Land Information System (GLIS)

Harnessing GLIS Data for Societal Challenges: Develop analytics solutions utilizing GLIS data to address societal challenges and inform evidence-based decision-making. Solutions should focus on domains like Urban Planning, Infrastructure Development, Environmental Conservation, Land Management, and Socio-economic Analysis.

II. Blockchain & Cybersecurity

1. Developing a system for generating and validating certificates using blockchain technology for government entities

Blockchain-Based Certificate Generation and Validation: Create a system for custom digital certificate generation and validation using open-source software and blockchain technology. Expected output includes certificate integration with a Digital Locker System, catering to Government Offices, Students, Industries, and Institutes.

2. Monitoring and ensuring compliance with security measures in cloud environments

Cloud Environment Monitoring and Auditing Tools: Develop software tools for monitoring and auditing cloud environments to detect security incidents, compliance violations, and anomalous activities. Features should include log analysis, intrusion detection, and security event correlation.

3. Creating an intelligent system utilizing AI/ML to identify phishing domains mimicking genuine websites

AI/ML-Based Phishing Domain Detection: Implement an automated tool utilizing AI/ML to identify phishing domains from genuine ones based on open-source databases like WHOIS. Techniques include code/content similarity analysis and web page image analysis for probability scoring of phishing domains.

III. Clean & Green Technology

1. Engaging students in innovative projects and initiatives

Waste Management and Sanitization Solutions: Propose solutions for waste segregation, disposal, and sanitation system improvements.

2. Developing a mobile application aimed at promoting environmentally friendly practices

Pro Planet Mobile Application: Design a mobile application to track user activities, promote eco-friendly behaviors, and suggest sustainable alternatives in real-time across various applications like food delivery and online ordering.

IV. Disaster Management

1. Creating a mobile app that gathers and displays water-related issues within communities through crowdsourcing and mapping

Social Media Data Utilization in Disaster Management: Develop software/algorithms to analyze ground-level images shared on social media for disaster management, focusing on water-related problems like flooding and drainage issues.

2. Establishing an application where rescue agencies register and coordinate their efforts, particularly during disasters

Rescue Agency Coordination Application: Create a centralized database and application for registering and locating rescue agencies to facilitate coordination and aid distribution during natural or man-made disasters.

V. FinTech

1. Exploring platforms for decentralized finance, potentially disrupting traditional financial systems

Decentralized Finance (DeFi) Platforms: Build decentralized finance applications leveraging cryptocurrency and blockchain for innovative financial services.

2. Predicting trends in various markets using data analysis and forecasting techniques

Market Trend Prediction Tools: Analyze historical and real-time market data to predict trends and optimize investment decisions.

VI. MedTech / BioTech / HealthTech

1. Designing software to aid in suggesting drugs and formulations based on Ayurvedic classical texts and repositories

Speech Therapy/Education Software for Neurodivergent Individuals: Develop software for speech therapy or education tailored for neurodivergent individuals.

2. Developing software to recommend medications and formulations for specific diseases or pharmacological properties based on Ayurvedic classical literature and repositories.

Ayurvedic Drug and Formulation Identification Software: Create custom software to identify Ayurvedic drugs and formulations tailored to specific symptoms and conditions, including pharmacological properties and contraindications.

VII. Tourism

1. Digital Tourism Solutions for Ayodhya

Digital Tourism Enhancement Solutions for Ayodhya: Leverage digital technologies like mobile apps, VR, AR, and geospatial mapping to enhance tourism experiences, promote Ayodhya as a tourist destination, and facilitate online bookings and navigation.

2. Develop a software application that leverages advanced technology to address key challenges and enhance the tourism experience in India. The application should provide innovative solutions to improve accessibility, promote cultural immersion, and streamline travel logistics for both domestic and international tourists.

"India boasts a rich cultural heritage, diverse landscapes, and a plethora of tourist attractions, making it a prime destination for travelers worldwide. However, the tourism industry in India faces various challenges, ranging from infrastructure limitations to language barriers and information accessibility.

To address these challenges, there is a need for a cutting-edge software application that harnesses technology to revolutionize the tourism sector in India. The application should incorporate features such as augmented reality (AR) for virtual tours of historical sites and landmarks, real-time language translation for seamless communication with locals, and personalized trip planning based on user preferences and interests.

Furthermore, the software should integrate with existing tourism databases and platforms to provide accurate information on accommodations, transportation options, local events, and cultural experiences. It should also offer safety and emergency assistance features to ensure the well-being of tourists during their travels.

The goal of this software application is to enhance the overall tourism experience in India by leveraging technology to overcome barriers, promote cultural exchange, and empower travelers to explore the country's diverse attractions with ease and confidence. Through innovation and digitalization, the application aims to position India as a leading destination for immersive and unforgettable travel experiences."

VIII. Miscellaneous

1. Creating immersive augmented reality (AR) experiences for various applications

Augmented Reality (AR) Application Development: Explore AR technology by developing Android apps overlaying digital content onto the physical world, fostering innovation and enhancing user experiences.

2. Ensuring the robustness and reliability of algorithms through rigorous testing and validation processes

Robust Algorithm Design: Design algorithms resilient to outliers, noise, missing data, and adversarial attacks, emphasizing robust optimization and statistics techniques.